

connected to the internet via ngrok-exposed FastAPI endpoints.

Version control and source management: All code artifacts, including backend services, scraping scripts, and mobile source code, were hosted on GitHub. The repository (<https://github.com/sakina27/price-scanner-app>) enabled version control, issue tracking, and team collaboration. GitHub was also integrated with Jenkins to trigger automated build pipelines on every push to the main branch.

Scraper execution: Each scraping service was encapsulated in its own Docker container and configured to run in headless mode using ChromeDriver. Execution was managed via cron jobs or manual triggers. Selenium waits ensured dynamic elements were fully loaded, while Chrome DevTools Protocol was employed to intercept and extract clean JSON payloads from API calls.

Figure 2 shows a formatted JSON output containing product data scraped from multiple online grocery platforms. Each entry includes the item name, price, store source, image URL, and

product link used by the system for real-time price comparison.

Mobile integration testing: The Android app was tested with live data sourced from the backend via ngrok URLs. Functional tests covered UI rendering, JSON parsing, error handling, and image loading. The app successfully displayed over 100 unique products per store, with no significant lags or crashes. Figure 3 shows the mobile UI of the grocery price comparison app. Upon searching for ‘milk’, the app displays product listings from various online retailers like Zepto, StarQuik, and JioMart, along with prices, images, and actionable buttons for purchase or comparison.

CI/CD pipeline validation: A multi-stage CI/CD pipeline was implemented using Jenkins, triggered automatically on each commit to the GitHub repository. The pipeline consisted of the following stages:

- Building the Android APK using Gradle.
- Creating Docker images for the scraper and API services.

- Pushing the images to Docker Hub.
- Deploying the services to a Kubernetes cluster using declarative YAML manifests.

To ensure safe and incremental rollouts, the pipeline was tested with canary deployments and rollback procedures. Figure 4 shows the Jenkins Stage View for the full pipeline. It visually represents each phase from source checkout and APK generation to image building, Docker Hub publishing, and Kubernetes deployment, along with the time taken for each step, offering insight into overall performance and pipeline efficiency.

Containerisation and orchestration: All services, including scraping bots, backend APIs, and auxiliary tools were containerised using Docker. Docker is a platform that enables developers to package applications along with their dependencies and environment configurations into lightweight, portable containers. This ensures consistency across different stages of development and deployment, and eliminates the classic ‘it works on my machine’ problem.

```

Pretty-print
[
  {
    "Name": "Modern Cremtreat Chocolate Cream Bun",
    "Price": 10,
    "Original Price": 274,
    "Image URL": "https://cdn.zeptonow.com/production/ik-seo/tr:w-400,ar-2400-2400,p-true,f-auto,q-80/cms/product_variant/238096ba-698f-4f5d-81f7-ff9e99869f4/Modern-Cremtreat-Chocolate-Cream-Bun.jpeg",
    "Product URL": "https://www.zepto.com/pn/modern-cremtreat-chocolate-cream-bun/pvid/524bc3f4-92e1-42e2-b96e-7a52bde947a2",
    "Source": "Zepto"
  },
  {
    "Name": "Modern Danish Bun",
    "Price": 10,
    "Quantity": "60 Gm",
    "Image URL": "https://media.starquik.com/catalog/product/SQ165192.jpg",
    "Product URL": "https://www.starquik.com/product/modern-danish-bun-60-gm",
    "Source": "StarQuik"
  },
  {
    "Name": "Wibs Mini Bread 200 g",
    "Price": 18,
    "Original Price": "10% OFF",
    "Image URL": "https://www.jiomart.com/images/product/original/490626262/wibs-mini-bread-200-g-product-images-o490626262-p608355598-0-202403050640.jpg?im=Resize=(360,360)",
    "Product URL": "https://www.jiomart.com/p/groceries/wibs-mini-bread-200-g/608355598",
    "Source": "Jio-Mart"
  },
  {
    "Name": "Repose Slice Bread - Premium White, Wholesome, Fresh, No Preservatives",
    "Price": 18,
    "Original Price": null,
    "Image URL": null,
    "Product URL": "https://www.bigbasket.com/pd/40251380/repose-slice-bread-premium-white-wholesome-fresh-no-preservatives-200-g-pouch/?nc=cl-prod-list&t_pos_sec=1&t_pos_item=40&t_s=Slice%2520Bread%2520-%2520Premium%2520White%2520C%2520Wholesome%2520Fresh%2520No%2520Preservatives",
    "Source": "Big Basket"
  },
  {
    "Name": "Bonn Bread Soft White Bread - Sliced, No Trans Fat",
    "Price": 18.8,
    "Original Price": null,
    "Image URL": null,
    "Product URL": "https://www.bigbasket.com/pd/40286394/bonn-soft-white-bread-premium-rich-great-taste-200-g/?nc=cl-prod-list&t_pos_sec=1&t_pos_item=37&t_s=Soft%2520White%2520Bread%2520-%2520Sliced%2520C%2520No%2520Trans%2520Fat",
    "Source": "Big Basket"
  },
  {
    "Name": "Kwality Burger Bun",

```

Figure 2: JSON response structure from product listing API showcasing data format